



Master Specification: Dispatcher Module

Role Tier: Level 2 (Operational Coordinator)

Reporting To: Logistics Manager (Owner)

Primary Objective: To maximize "Vehicle Utilization," minimize "Cost per Mile," and ensure "On-Time Delivery" (OTD).

1. The Dispatch Control Center (Dashboard)

The Dispatcher needs a high-intensity dashboard focused on movement and assignment.

- **Pending Load Queue:** A list of "Ready-for-Dispatch" manifests coming from the **Warehouse Manager**.
- **Active Driver Roster:** A live list of available drivers, their current location, their remaining "Hours of Service" (safety compliance), and their vehicle's current load capacity.
- **Map-Centric Interface:** A Google Maps view showing "Unassigned Orders" (Grey dots) vs. "Active Routes" (Colored lines).

2. Dynamic Route Optimization

The Dispatcher doesn't just pick a path; they solve a "Traveling Salesman Problem" (TSP) in real-time.

- **Automated Batching:** The system groups orders based on geographic clusters to ensure a driver isn't driving across town for a single package.
- **Constraint-Based Routing:** The system calculates routes based on:
 - **Traffic Data:** Real-time Google Maps traffic pings.
 - **Vehicle Constraints:** Ensuring a large truck isn't sent down a narrow residential street or a low-clearance bridge.
 - **Priority Windows:** Ensuring "Express" orders are at the start of the route.
- **Manual Override:** The Dispatcher can "drag and drop" an order from one driver to another on the map if an emergency occurs.

3. Driver Dispatch & Hand-off

- **Digital Manifest Push:** Once the Dispatcher hits "Confirm," the route is pushed directly to the **Driver's** mobile app.
- **Load Balancing:** Ensuring no single driver is over-burdened while others are idle.
- **ETA Calculation:** The system provides an "Estimated Time of Arrival" for every stop on the route, which is then shared with **Customer Support**.

4. Real-Time Crisis Management

The Dispatcher is the first responder to system disruptions.

- **Re-Routing:** If a truck breaks down or a road is closed, the Dispatcher can "Re-optimize" the remaining stops and push the update to the driver instantly.
- **Geofence Alerts:** Receives alerts if a driver deviates from the assigned route or enters a "No-Go Zone" defined by the **Logistics Manager**.
- **Communication Hub:** A direct chat/voice link to both the **Warehouse Manager** (to ask about a missing box) and the **Driver** (to ask about a delay).

5. Generative AI Integration (The "Smart Dispatcher")

How AI turns a Dispatcher into a Super-Coordinator:

- **AI Route Explanation:** Instead of just seeing a line, the AI explains: *"I've shifted Route 4 to avoid the 3 PM stadium traffic; this saves 12 minutes and 1.5 gallons of fuel."*
- **Proactive Delay Prediction:** The AI analyzes weather and traffic patterns and warns: *"There is a 70% chance Driver B will miss their last 3 windows due to rain. Should I move those orders to Driver C now?"*
- **Natural Language Assignment:** The Dispatcher can type: *"Assign all small parcels in the downtown area to our most fuel-efficient van,"* and the system executes the command.

6. Performance & Metrics

The Dispatcher is measured by the efficiency of their "Plans":

- **Empty Miles:** Tracking how much a truck drives without a load (Goal: 0).
- **SLA Compliance:** Percentage of orders assigned and picked up within the agreed timeframe.
- **Route Deviation Rate:** How often drivers have to ignore the Dispatcher's plan (indicating a poor plan).

Technical Hand-shake for your Team:

From Role	To Dispatcher	Data Transferred
Warehouse Manager	Manifest Data	Total boxes, weight, and "Ready" timestamp.

Logistics Manager	Driver Credentials	Who is authorized to drive today.
Driver	GPS Telematics	Real-time coordinates and fuel levels.
